



Heat protection plan Vorarlberg 2020

imprint

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Foreword

Many people are happy about beautiful, sunny summer days - but heat also poses dangers. More and more hot days and tropical nights therefore require more and more intensive climate protection measures and active adaptations to protect us from them.

This heat protection plan is aimed at the Vorarlberg population and system partners such as municipalities, nursing homes or childcare facilities. The information is intended to help identify potential health risks and demonstrate effective countermeasures.

This includes advice on precautionary measures, suggestions for urban development measures, an early warning with a precise communication chain and information for those affected on the measures they can take in response to the warning.

The Vorarlberg Heat Protection Plan is thus, for the first time, a comprehensive information brochure that will also serve as a basis for further developments in this area in the coming years.




Dipl.-Vw. Andrea
Kaufmann Mayor
Vice President Municipal Association




Mag. Markus Wallner
Governor

1. Heatwaves as a challenge

1.1. Heatwaves

Due to global warming, more frequent and more severe heat waves and extreme precipitation events are physically to be expected. Recent heat waves in Europe have led to an increase in associated mortality, but the adverse health effects of hot weather and heat waves are largely preventable. Hot days are days with a maximum temperature of at least 30°C. A heat wave is defined (according to the Czech meteorologist Jan Kysely) essentially as a series of at least three consecutive days above 30°C. Temperature extremes have changed significantly, for example cold nights have become rarer, but hot days have become more frequent. In the 21st century, this development will continue intensified and the frequency of heat waves will also increase." [Austrian Climate Change Assessment Report 2014 - Summary for Decision-Makers]

1.2. Heat warning from ZAMG

The ZAMG warns of extreme weather events nationwide. The intensity of a predicted event, and thus the extent of the expected impacts, are characterized by warning colors. The warning colors yellow, orange and red are used for heat stress, which stand for caution, attention and danger respectively. Heat warnings are created based on forecast values for the "perceived temperature" (air temperature, humidity, wind and indirectly also radiation).



www.zamg.ac.at

1.3. Heat warning from AGES

In the event of prolonged or particularly severe heat stress, the Ministry of Health, in cooperation with the Austrian Agency for Health and Food Safety (AGES), activates a heat hotline and provides practical tips and advice to the population free of charge.



www.ages.at

1.4. Heat warning from the state of Vorarlberg

The national heat protection plan stipulates that as soon as the forecast calculations of the Central Institute for Meteorology and Geodynamics (ZAMG) predict heat stress, an automatic heat warning is sent to the affected federal states, the Ministry of Health and the Chamber of Pharmacists. An expansion of the weather service on the "Vorarlberg Weather Box" set up between the state of Vorarlberg and the ZAMG is being planned and stipulates that the ZAMG will provide additional information on the subject of "heat" if an orange or red heat warning is issued on the "ZAMG homepage" for Vorarlberg for at least three days in a row.

When warnings are received, the states provide specific facilities such as retirement homes, hospitals and kindergartens, emergency organizations and the population with specific information.

The state of Vorarlberg has been operating its own warning website for two years to alert the population, and the latest warning information is also distributed via Facebook, Twitter and Instagram and the KATWARN app. In addition, the Vorarlberg Environmental Institute informs the population via the Internet and the ozone telephone, along with health-related recommendations for behavior, as soon as the ozone information threshold is exceeded.



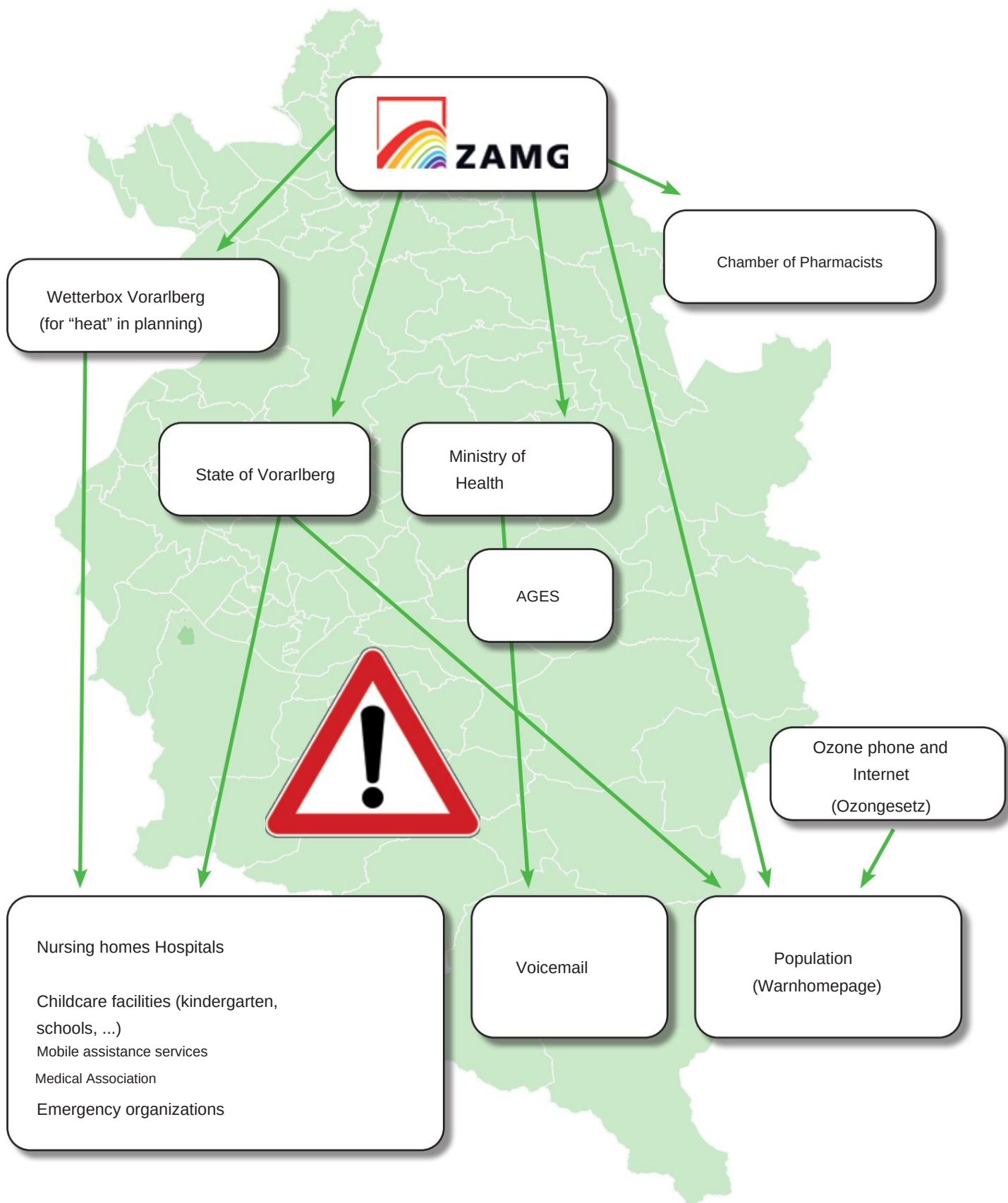
www.vorarlberg.at/warnung

[Vorarlberg State Warning Center](#)

[@lwzvorarlberg](#)

[lwzvorarlberg](#)

Heat warnings for Vorarlberg



1.5. Measures at community level

The establishment of public social spaces that people can use to cool down temporarily is primarily the responsibility of the municipalities. However, the state of Vorarlberg is providing financial support for planning.

Countermeasures to heat stress by municipalities

- Preservation and maintenance of inner-city greenery •

Creation of shaded outdoor spaces (such as playgrounds)

- Operation of drinking water fountains in public spaces

- ecological improvement of inner-city watercourses • high

standards in the construction of public buildings in terms of overheating protection

- Raising awareness of correct behaviour in kindergartens and schools
- Minimizing soil sealing and expanding green areas (e.g. green roofs)
- Further reduction of air pollutants, especially ozone precursors
- Raising awareness to minimize heat input through optimized building orientation and shading for south-facing windows

As an incentive to implement measures for the shading and greening of buildings when constructing municipal buildings, e.g. for green facades, infiltration areas or green roofs, additional points on the municipal building permit (KGA) and thus increased state funding have been possible since 2016. The amendment to the Spatial Planning Act in 2019 also stipulated that the spatial development plan must take the requirements of climate change into account as a central aspect when determining the desired settlement development.

The federal government and the state government office provide municipalities and businesses with subsidized advice on natural outdoor spaces (green roofs and greenery around the building) as part of the Impuls3 advisory program.

A brochure and a poster collection have been compiled on the possible fields of action for climate change adaptation at the municipal level, which also provides information on the topics of public green spaces, building cooling, shading, etc. www.vorarlberg.at/klimawandelanpassung

In the area of climate change adaptation at the municipal level, the state of Vorarlberg promotes orientation discussions through climate change adaptation consultations from the Vorarlberg Energy Institute, as well as detailed risk and climate impact analyses. In addition, there is the Climate Change - Adaptation Model Regions (KLAR!) funding program of the Climate and Energy Fund and corresponding support options from the state for this.



2. Specific geographical conditions

The ZAMG operates numerous measuring stations in Vorarlberg; the data can be viewed on their homepage and on the ORF weather page. In addition to various air quality parameters, the state's air quality measuring stations also continuously record meteorological data such as air temperature and transfer these online to the Environmental Institute's measuring network center.

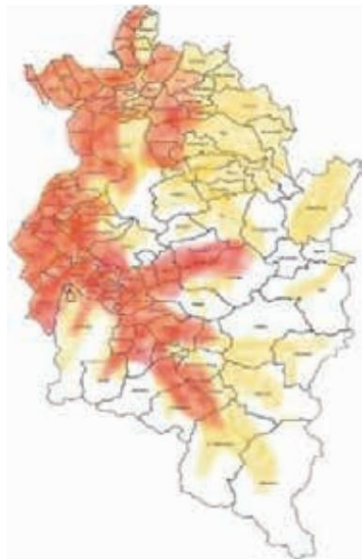
Due to the particular heat and special circumstances of the population with regard to old and young, preventive measures in general represent a particular challenge for future planning, especially for the municipalities of Vorarlberg.

Analysis: Communities relatively more affected by heat

2.1. Procedure

In a first step, the places where heat stress is highest were selected: hot days and tropical nights. These places were analyzed in terms of their age structure and other socio-demographic aspects (number/proportion of children and elderly people, old age dependency ratio, single households 65+).

From this, three area types were formed.



The places where the average daily maximum temperature between 1981 and 2010 was between 0 and 7 days above or equal to 30°C (= hot days) are marked in yellow. The regions where the average during this period was = 0 are marked in white.

Min.: 0 Takes Max.: 6.4 Takes

The locations marked in red are those where hot days were recorded between 1981 and 2010 and where the average was between 0 and 3 nights in which the daily minimum temperature was greater than or equal to 20°C (= tropical nights).

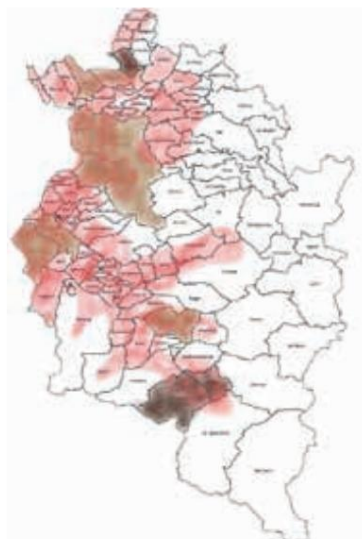
Min.: 0 nights Max.: 1.6 nights

During this period, there were no places in Vorarlberg where tropical nights but no hot days were recorded.

Additionally, it should be noted that heat does not take municipal boundaries into account.

Quelle: (ZAMG, Hiebl et al. 2015) (ZAMG, Hofstätter et al. 2016) (Uni Graz, Wegener Center, Leuprecht et al. 2016), State Office for Statistics (VWZ 2019); own representations

2.2. Heat in connection with socio-demographic characteristics



The map shows in red the places with **hot days and tropical nights**.

Municipalities in which there is a high **number of children** under 10 years of age, a high **number of people** aged 65 and over, and a high number of **single households** with one person aged 65+ have been marked in **light brown**.

The municipalities with a high proportion of people aged 65 and over and a high proportion of

Single households with one person aged 65+.

Source: own illustration

2.3. Typifications

Type A:

Communities with a high number of children under 10 years of age, people aged 65 and over and single households with one person aged 65+.

Bludenz	628 children under 10 years	2,767 people 65+	956 single households 65+
Bregenz	1,269 children under 10 years	5,722 people 65+	2,085 single households 65+
Dornbirn	2,171 children under 10 years	8,780 people 65+	2,671 single households 65+
Feldkirch	1,564 children under 10 years	5,873 people 65+	1,714 single households 65+
in Götz	523 children under 10 years	2,001 people 65+	511 single households 65+
Hard	546 children under 10 years	2,411 people 65+	675 single households 65+
Hohenems	802 children under 10 years	2,829 people 65+	734 single households 65+
Lauterach	431 children under 10 years	1,644 people 65+	450 single households 65+
Lustenau	1,102 children under 10 years	3,989 people 65+	1,132 single households 65+
Rankweil	497 children under 10 years	2,229 people 65+	584 single households 65+
in total	9,533 children under 10 years	38,245 people 65+	11,512 single households 65+

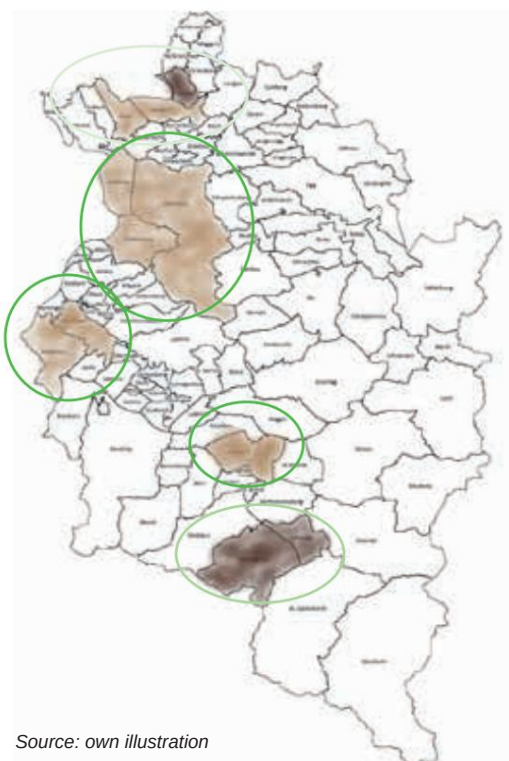
TYPE B:

Communities with a high proportion of persons aged 65 and over in the population in the communities and a high proportion of single households with one person aged 65+ in relation to all private households in the community.

Lochau	1,444 people 65+ (24% of the population)	442 single households 65+ (16% of households)
Schrüns	903 people 65+ (23% of the population)	297 single households 65+ (17% of households)
Tschagguns	556 people 65+ (21% of the population)	138 single households 65+ (15% of households)
total	2,903 people 65+	877 single households 65+

Source: State Office for Statistics (VWZ 2019)

2.4. Description of the relatively more affected communities



In the following map, the community types from the upper map have been grouped into area types.

Area type A: Places with municipalities with a high number of children under ten years of age, a high number of people aged 65 and over and a high number of single households with one person aged 65+.

Area type B: Places with municipalities with a high Proportion of persons aged 65 and over in of the population in the municipalities and a high proportion of single households with one person aged 65+ among all private households.

Area type C: Places with municipalities with characteristics of type A and type B.

2.5. Hot days and climate change

While hot days with temperatures above 30 °C have so far been very rare in Vorarlberg, hot days will become the norm for many people in the future due to global warming. Even if the Paris climate targets are achieved, there will be only a few communities without hot days. In 21 communities there was no hot day on average between 1971 and 2000, meaning it was never hotter than 30 degrees. Nine communities had four hot days (source: Addendum / ZAMG). Due to climate change, hot days are increasing in Vorarlberg. In the worst case scenario, there will be an average of 17 to 41 hot days per summer between 2071 and 2100. Communities in low-lying areas are particularly affected. While no significant increase in hot days is expected in high-lying areas in the near future, hot days are also expected in high-lying areas by the end of the century.

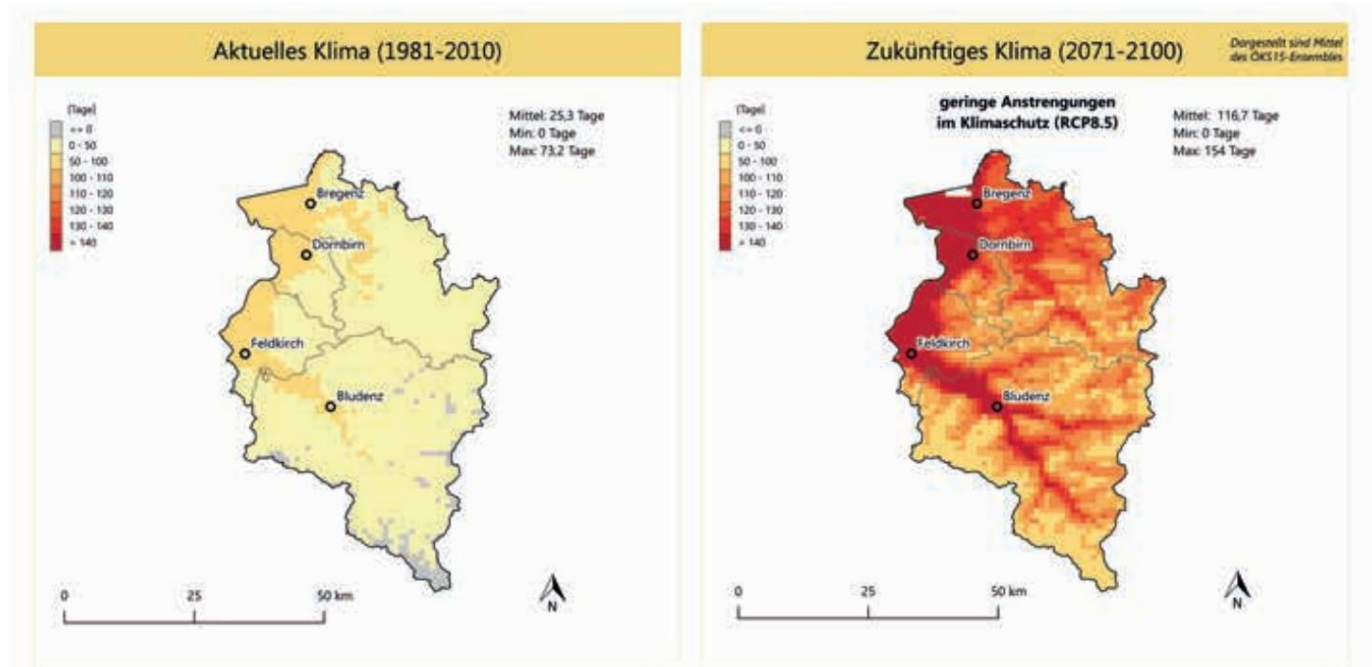


Figure 1 – Development of hot days (daily temperature > 30 °C) (Source: Prutsch A. et.al. (2018): Climate impact maps for Austria's regions Background information for use. A brochure from the research project CLIMA-MAP. alpS, Innsbruck.)

Development of hot days in Vorarlberg from 1971 to 2100

Community code	Name	 Hot days earlier (1971-2000) 4 4	 talk Paris scenario (2021-2050)	 talk Climate protection scenario (2021-2050)	 talk Business-as-usual scenario (2021-2050)	 talk Paris scenario (2071-2100)	 talk Climate protection scenario (2071-2100)	 talk Business-as-usual scenario (2071-2100)
80404 Feldkirch		4	6-13	7-11	6-13	6-11	8-18	15-36
80401 Altach		4	6-13	7-11	7-12	6-11	8-17	14-35
80412 Mäder			5-13	7-12	7-14	6-11	9-19	16-39
80408 Götz			5-13	7-11	7-13	6-11	8-18	15-36
80405 Frastanz	4	5-13	6-10	6-11	6-11	5-11	7-16	13-33
80302 Hohenems	4	6-13	7-11	7-12	7-12	6-11	8-17	14-35
80409 Klaus	4	6-13	6-11	6-12	6-12	5-11	8-17	14-35
80423 Weiler	4	6-13	6-11	6-12	6-12	5-11	8-17	14-35
80413 Meiningen	4	6-13	8-13	8-15	8-15	6-12	10-20	17-41
80418 Schlins	3	4-12	5-9	5-10	5-10	4-10	6-15	11-31
80303 Lustenau	3	4-13	7-12	7-13	7-13	4-10	9-19	16-40
80103 Bludenz	3	5-13	5-8	5-10	5-10	5-11	6-16	10-34
80410 Koblach	3	4-11	7-12	7-14	7-14	4-9	9-19	16-39
80240 Wolfurt	3	5-12	5-10	5-10	5-10	4-9	7-16	13-35
80224 Lauterach	3	5-13	6-10	5-11	5-11	5-10	7-16	14-35
80217 Höchst	3	5-13	7-13	7-13	7-13	4-10	9-19	16-40
80117 Nüzider	3	5-13	5-9	5-10	5-10	5-11	6-17	11-35
80416 Røthis	3	4-11	5-9	5-10	5-10	4-9	6-15	12-32
80214 Gailåu	3	4-12	7-13	7-13	7-13	4-10	9-19	16-40
80420 Sulz	3	4-11	5-9	5-10	5-10	4-9	6-15	12-32
80104 Bludesch	3	5-12	5-9	5-11	5-11	4-10	6-16	11-32
80213 Fußbach	3	5-13	7-13	7-13	7-13	5-10	9-19	16-40
80114 Lorüns	3	4-15	4-8	4-10	4-10	5-12	5-16	9-34
80407 Göfis	3	4-11	6-10	5-11	5-11	4-9	7-16	13-33
80414 Rankweil	3	4-11	5-10	5-11	5-11	4-9	7-16	13-34
80126 Thuringia	3	5-12	5-9	5-10	5-10	4-9	6-15	10-32
80415 Røns	3	4-11	5-9	5-10	5-10	4-9	6-15	11-31
80226 Lochau	3	4-12	6-10	5-10	5-10	5-9	7-15	13-34
80215 Hard	3	5-13	7-12	6-12	6-12	5-10	8-18	16-39
80417 Satteins	3	5-11	5-9	5-11	5-11	4-9	6-15	12-32
80207 Bregenz	2	4-11	5-10	5-10	5-10	3-8	6-15	13-34
80301 Dornbirn	2	3-9	4-8	4-9	4-9	3-6	5-14	11-31
80211 Egg	2	3-9	3-6	3-7	3-7	3-6	4-12	9-28
80128 Tschagguns	2	3-12	2-5	2-6	2-6	3-9	3-12	6-29
80129 The problem	2	3-13	3-5	2-7	2-7	3-10	3-13	7-31
80220 Kennelbach	2	3-9	5-9	4-10	4-10	3-6	6-15	13-33
80210 Doren	2	3-8	3-7	3-8	3-8	3-6	4-12	11-29
80219 Hohenweiler	2	3-9	4-8	4-9	4-9	3-7	5-13	12-32
80218 Hörbranz	2	4-10	5-9	4-9	4-9	3-7	6-14	12-33
80424 Zwischenwasser	1	2-6	3-6	3-8	3-8	2-4	4-12	9-28
80123 Silbertal	1	2-7	1-3	1-5	1-5	2-5	2-10	4-28
80233 Schoppernau	1	2-7	1-3	1-4	1-4	2-5	1-8	3-24
80202 Andelsbuch	1	2-8	3-6	3-7	3-7	2-5	4-12	9-28
80232 Schnepfau	1	2-6	1-4	1-5	1-5	2-4	2-10	5-26
80206 Bizau	1	2-6	2-5	2-6	2-6	2-5	3-11	7-27
80212 Eichenberg	1	2-6	3-7	3-8	3-8	2-4	4-13	11-31
80111 Innerbraz	1	2-7	1-4	1-5	1-5	2-5	2-10	5-27
80122 Schruns	1	2-8	1-4	1-6	1-6	2-6	2-11	5-30
80116 Nenzing	1	2-6	2-5	2-6	2-6	2-4	3-10	7-24
80230 Reuthe	1	2-7	2-5	2-6	2-6	2-5	3-11	7-26
80124 Sunday	1	2-7	1-3	1-3	1-3	2-5	1-6	3-21
80107 Bürserberg	1	2-7	2-6	2-7	2-7	2-5	3-12	7-30
80419 Schnifis	1	2-7	3-7	3-8	3-8	2-5	4-13	9-29
80115 Ludesch	1	2-8	3-7	3-8	3-8	2-5	4-13	8-30
80106 Bürs	1	2-7	2-6	2-7	2-7	2-5	3-13	7-31
80127 Thüringerberg	1	2-5	2-5	2-6	2-6	2-4	3-11	6-26
80221 Krumbach	1	2-7	3-6	2-7	2-7	2-5	3-11	9-27
80108 Dalfas	1	2-9	1-3	1-4	1-4	2-7	1-8	3-23
80225 Lingenau	1	2-6	2-5	2-7	2-7	2-4	3-11	9-27
80125 Stallehr	1	2-8	2-6	2-7	2-7	2-6	3-13	7-32
80223 Langenegg	1	2-6	2-6	2-7	2-7	2-4	3-12	10-28
80121 Sankt Gerold	1	2-7	2-5	2-6	2-6	2-5	3-10	6-25
80222 Langen near Bregenz	1	2-7	3-7	3-8	3-8	2-5	4-12	10-30
80216 Hittisau	1	2-6	2-5	2-6	2-6	2-4	3-11	8-26
80235 Schwarzach	1	2-7	4-8	4-9	4-9	2-5	5-14	12-32
80208 Book	1	2-6	3-7	3-8	3-8	2-4	4-12	10-30
80203 Au	1	2-6	1-4	1-4	1-4	2-5	2-9	4-25
80422 Viktorsberg	1	2-5	3-6	3-8	3-8	2-4	4-12	9-28
80205 Bildstein	1	2-7	4-8	4-9	4-9	2-5	5-14	12-33
80201 Alberschwende	1	2-6	3-6	2-7	2-7	2-4	4-12	10-29
80119 Sankt Anton im Montafon	1	2-8	2-5	2-7	2-7	2-6	3-13	7-32
80402 Duns	1	2-6	3-7	3-8	3-8	2-4	4-13	9-28
80227 Mellau	1	2-7	2-4	2-5	2-5	2-5	3-9	6-23
80236 Schwarzenberg	1	2-7	2-6	2-7	2-7	2-5	3-11	9-27
80204 Bezaú	1	2-7	2-5	2-6	2-6	2-5	3-11	7-27
80421 Übersaxen	1	1-3	2-6	2-8	2-8	0-2	4-12	9-29
80112 Klostertal	1	0-5	0-1	0-2	0-2	0-4	0-4	1-17
80231 Riefensberg	1	1-4	2-5	2-6	2-6	1-2	3-10	8-25
80234 Schröcken	1	0-5	0-2	0-2	0-2	0-4	0-7	1-24
80113 Lech	0	0-6	0-1	0-2	0-2	0-6	0-6	1-20
80239 Warth	0	0-3	0-1	0-2	0-2	0-2	0-6	1-21
80406 Fraxern	0	1-4	3-6	3-8	3-8	1-2	4-12	9-29
80238 Sulzberg	0	1-4	2-5	2-6	2-6	0-2	3-11	9-28
80237 Sibratsgfall	0	1-3	1-3	1-4	1-4	0-2	2-8	4-21
80120 Sankt Gallenkirch	0	0-6	1-2	1-3	1-3	0-5	1-7	2-24
80403 Dünserberg	0	0-2	2-6	2-7	2-7	0-1	3-12	8-27
80118 Raggal	0	0-3	1-4	1-5	1-5	0-2	2-10	5-26
80209 Damüls	0	0-0	0-2	0-2	0-2	0-0	1-5	2-19
80229 Möggers	0	1-4	2-6	2-7	2-7	0-3	3-11	9-29
80101 Bartholomäberg	0	0-4	1-4	1-6	1-6	0-2	2-12	5-32
80102 Blondes	0	1-3	1-3	1-4	1-4	0-2	2-8	4-23
80110 Gaschurn	0	1-8	1-2	1-2	1-2	0-7	1-5	2-20
80109 Drinking fountain	0	1-4	1-2	1-3	1-3	0-3	1-6	2-21
80105 Brand	0	1-5	1-2	1-3	1-3	0-3	1-5	3-17
80228 Mittelberg	0	1-3	0-2	0-2	0-2	0-2	1-5	2-19
80411 Laterns	0	1-3	2-5	2-6	2-6	0-2	3-10	7-24

Source: ZAMG/ Addendum, own illustration

3. Heat and health

The influence of meteorological conditions on human health is well known and has been the subject of numerous scientific studies. For heat stress, such studies show, among other things, increased mortality and diseases of the cardiovascular system, kidneys, respiratory tract and metabolic disorders.

3.1. Interaction between heat and air pollution

Air is a gas mixture that consists of the main components nitrogen and oxygen. There are also noble gases and other trace substances such as carbon dioxide, exhaust gases, etc. Air pollutants, such as those released during various activities, affect the health of humans and animals, but are also harmful to vegetation, soil and water. The lungs are most affected by air pollution. However, air pollutants also damage other organ systems and worsen chronic diseases. If air pollutants such as fine dust enter the human body via the lungs, they can cause a number of diseases. Breathing air contaminated with pollutants during heat waves is an additional health risk factor. Summer heat has an insignificant effect on air pollution caused by fine dust and nitrogen dioxide. However, it does contribute to increased ozone formation. The mixture of a large number of photooxidants that forms from various precursor substances near the ground under the influence of strong sunlight is called summer smog. The precursor pollutants primarily include nitrogen oxides and volatile organic hydrocarbons. The main substance causing summer smog is ozone.

Ozone irritates the mucous membranes (burning eyes, scratchy throat, pressure on the chest and pain when breathing deeply), can cause inflammation of the respiratory tract and lungs, temporarily limit lung function, impair physical performance and cause breathing difficulties. It also increases the reaction to other air pollutants and allergy triggers such as pollen and mites. High ozone levels can build up over the course of hot days, particularly when there is no wind. Ozone levels are highest in the afternoon. During the night, levels decrease again. During longer periods of stable good weather in spring and summer, ozone levels can build up continuously over days and reach high peak levels. It is important to remember that in addition to ozone, the precursor pollutants themselves and other pollutants, especially those caused by traffic, are also substances that are relevant to health. At higher concentrations, these pollutants are at least as problematic as ozone.

Since the beginning of the 1990s, information campaigns to raise awareness of air pollution issues and calls for voluntary measures in the event of increased ozone pollution have been part of the Vorarlberg Ozone Programme.

In addition, various efforts have been made to reduce air pollution and environmental initiatives have been launched. Up until 2014, peak ozone concentrations have decreased significantly compared to previous years. After 2014, peak pollution levels tended to increase again slightly.

Reasons for the decline include the success in reducing emissions of nitrogen oxides (from industrial plants and road traffic, among others) and volatile hydrocarbons as precursors of ozone. Since emissions of volatile hydrocarbons will not continue to fall in the coming years and will in fact increase again, it must be expected that ozone pollution will rise again, together with global warming and the associated more frequent heat waves.

Based on these findings, it appears necessary to make every effort to keep the pollutant content in the air we breathe as low as possible so that this health risk factor is not added to heat stress. Emissions of precursor substances must therefore be reduced further than the measures already taken.

3.2. Information for

3.2.1. the population during heat waves

KEEP YOUR APARTMENT OR HOUSE COOL



• Keep room temperature below 32 °C during the day and below 24 °C at night • Open windows and shutters at night and early in the morning • Close windows and shutters that are exposed to the sun during the day • Install blinds, curtains, awnings or shutters on windows • Hang wet towels to cool the room air • In homes with air conditioning, close doors and windows • Fans can provide relief • Plant trees in your garden

GET OUT OF THE HEAT



• Go to the coolest room • Spend two to three hours during the day in a cool place (air-conditioned public building) • Do not go outside during the hottest time of the day • Do not do any physical exertion • Stay in the shade, go to the forest • Do not leave children or animals in parked vehicles • Take cold showers or baths • Wear light, loose-fitting clothing made of natural materials, and wear a hat and sunglasses when outdoors • Use light bedding and do not use heavy blankets

DRINKING IN HIGH HEAT



• Drink before you feel thirsty • Drink water or diluted fruit juice regularly and avoid alcohol • Make sure that children are regularly asked to drink water • Remind older people to drink and encourage them to do so

EATING IN HIGH HEAT



• Eat more often, but less and avoid high-protein foods • Eat less fatty foods and less meat • Offer children fresh, light food (e.g. melons, cucumbers, fruit)

HELP OTHERS



- Visit family members, friends and neighbours who are often alone
- Talk to your family about extreme heatwaves
- Provide advice and help to people at risk. Visit elderly people living alone at least once a day
- Ask the doctor how the prescribed medications affect body temperature and Can affect fluid balance
- Attend a first aid course and learn what to do in heat-related emergencies

IF YOU ARE IN ILL HEALTH



- Store medicines below 25 °C or in the refrigerator
- If you have a chronic illness or are taking medication, seek medical advice
- Sick children: Diarrhea leads to rapid loss of fluids. Fatigue, tiredness, Pay attention to dry skin and lips

IF YOU OR OTHERS FEEL UNWELL



- If you feel dizzy, weak, or have a strong thirst or headache, get help, drink a cool
- Visit the location and measure body temperature
- Drink water or fruit juice to compensate for fluid deficiency
- For painful muscle cramps (legs, arms, abdomen, after extensive exercise, hot weather), rest in a cool place and drink electrolyte rehydration solutions. If cramps persist, seek medical attention
- If unusual symptoms occur or persist, consult a doctor

While waiting for help, lay the person flat in a cool place with legs and hips slightly elevated. Remove outer clothing and cool with cool packs and constantly fan and spray the skin with water at 25 - 30 °C.

Measure body temperature. Do not give acetylsalicylic acid or paracetamol.

Place unconscious person in recovery position.

3.2.2. Particularly vulnerable persons

Heat waves and hot weather can kill and can also worsen existing health conditions. Health effects can occur in all age groups. However, some people are at higher risk of heat-related illness and death than others. Who is most at risk?



The older

Older people represent the largest group, with older people with dementia being particularly at risk. Ageing reduces heat tolerance: thirst is detected later, the sweat reaction is delayed and the number of sweat glands is reduced.

Regular monitoring, ensuring proper clothing, cool environment, adequate nutrition and adequate fluid intake improve their ability to cope with heat waves.



Infants and children

The metabolism of infants and children differs from that of adults.

They also rely on others to regulate their thermal environment and ensure adequate fluid intake.



People with chronic diseases

Reasons why people with chronic diseases are at increased risk during heat waves

At risk are:

- Cardiovascular diseases, as thermoregulation in the case of severe heat stress is a healthy Cardiovascular system requires
- Peripheral vascular disease, often caused by diabetes or arteriosclerosis as the blood supply to the skin can be difficult
- Diarrhoea or feverish illnesses, especially in children, and kidney or metabolic diseases, as these may be associated with excessive fluid loss and dehydration
- Chronic diseases that affect the number and/or function of sweat glands such as diabetes, scleroderma and cystic fibrosis
- An illness that forces someone to stay in bed and limits their ability to care for themselves or leave the house on a daily basis
- Some of the medications prescribed for these chronic diseases may themselves increase the risk of heat illness.



People taking certain medications

Many drugs can directly affect the thermoregulatory center, sweating, cutaneous vasodilation and/or increasing cardiac output and thus heat dissipation. For example, vasodilators such as nitrates and calcium channel blockers can theoretically cause low blood pressure in people who tend to become dehydrated with excessive heat exposure, especially the elderly.

Dehydration and changes in blood volume distribution can also increase the toxicity of drugs and/or reduce their efficacy by affecting drug levels, drug kinetics and excretion and thus pharmacological activity. Finally, storage of drugs at high ambient temperatures can affect their efficacy, as most manufactured drugs are approved for storage at temperatures up to 25 °C. This is particularly important for emergency drugs such as antibiotics, adrenaline, analgesics and sedatives.



People in certain professions

Anyone who has to work outdoors in hot weather without adequate protection, especially when it involves heavy physical exertion (e.g. roofers, field workers or road workers), is at increased risk of suffering health damage from heat and direct sunlight. Protective clothing, especially for emergency service workers, can be dangerous.

Further information on this topic can be found at: •

www.gesundheit.gv.at

• www.auva.at

• www.arbeitsinspektion.gv.at

3.2.3. Nursing homes

Line

- Lowering the indoor temperature in hot weather
- Room temperature between 08:00 and 10:00, at 13:00 and at night after 22:00
Check. The room should not be more than 32 °C during the day and 24 °C at night
- Open all windows at night and early in the morning to cool down the apartment
- Close all windows and shutters during the hottest time of day: heat outside and cool air
leave inside
- Install blinds, curtains, awnings or shutters on windows that are exposed to the morning or afternoon
Sun is shining
- Reduce heat stress indoors: switch off lights and electrical devices if possible,
but not the refrigerator
- Hang wet towels to cool down the room air
- Shading the windows from the outside reduces the amount of heat they get from the sun. The interior
Shading on the windows is advisable
- Fans provide relief, but if the temperature exceeds 35 °C, they may not prevent heat-related illnesses. They can
also cause excessive dehydration. It is advisable to place the fan at a certain distance and not point it
directly at the body, and to drink regularly. This is especially important for sick, bedridden people
- Use of mobile evaporative coolers: Their effect increases with temperature and decreases with
relative humidity
- Use of dehumidifiers: can be used in areas with high humidity, but not very high
Temperatures can be useful
- Use of local air conditioning: bring relief. Proper cleaning and maintenance is important to avoid health effects. Consider the
possibility of power outages in summer
- Inform and train staff

Personal

“Drink plenty of fluids” means consuming an amount of water to compensate for fluid losses (mainly through urine and sweat) of approximately 150%.

During heat waves and heatwaves, people should drink even if they do not feel thirsty. This is especially true for older people whose sense of thirst is impaired.

However, excessive drinking of pure water can lead to severe hyponatremia with complications such as stroke and death. The addition of sodium chloride and other suitable soluble substances to the drink (20 – 50 mmol/l) reduces water excretion through urine and facilitates the restoration of fluid balance.

- Follow the instructions for keeping rooms cool for the public (see point 2.2.1.) and ensure that patients and residents avoid the heat, keep cool and drink enough
- Monitor indoor temperatures. Provide a cool room (such as an air-conditioned room with a temperature below 25 °C) and move residents to this cool area for several hours each day
- Ensure medical review of the clinical care of vulnerable residents, if they suffer from chronic diseases
- Monitor residents' fluid intake. Non-alcoholic, unsweetened
Offer drinks
- Monitor body temperature, pulse, blood pressure and hydration
- Watch for early signs of heat-related illness and seek appropriate treatment if necessary.
Initiate treatment



3.2.4. Doctors

Act with foresight

- Regulate body temperature and hemodynamic responses in extreme heat
- Origin, clinical presentation, diagnosis and treatment of heat-related diseases
- Early signs of heat stroke are a medical emergency
- Appropriate cooling and resuscitation measures
- Disease risks and protective factors associated with heat waves
- Address at-risk patients with appropriate education about heat-related illnesses and how to prevent them
- Before the hot season, treat chronically ill patients with relevant
Combine with advice (avoid sun, drink fluids, medication). Due to possible side effects of the prescribed medication, adjust the dosage if necessary during heat and heat waves
- Medicines can lose their effectiveness at high temperatures and most medicines are only approved for storage up to 25 °C. Store and transport emergency medicines at an appropriate temperature
- Monitoring drug therapy and fluid intake, especially in elderly and frail
People and people with advanced heart disease

Provide patients with education, advice and information about

- How important it is to follow the instructions in the package leaflet
- How behavior, drug use and fluid intake are individualized depending on the clinical condition can be adjusted
- How to reach social and medical services, telephone information centres and emergency services

3.2.5. Healthcare professionals

Medications can affect the body's natural cooling and cause health problems in many ways
Way (WHO Regional Office for Europe, 2009) by:

- altered central temperature regulation and corresponding physiological and behavioral responses
- altered cognitive alertness accompanied by increased sleepiness and reduced heat avoidance
- altered blood pressure and cardiac output, vasodilation associated with impaired cooling, increased dizziness or fainting
- anticholinergic effects, blockade of the parasympathetic nervous system and inhibition of the natural Sweating for evaporative cooling
- altered renal function and electrolyte balance with increased risk of dehydration and drug toxicity or of overhydration and electrolyte imbalance.

DRUG

EFFECTS

Anticholinergic drugs	can inhibit central temperature regulation, cognitive Limit alertness, prevent or reduce sweating
Antipsychotics	can inhibit sweating, lower systolic blood pressure, inhibit central temperature regulation, limit cognitive alertness and vasodilation
Antihistamines	can inhibit sweating, lower systolic blood pressure
Anti-Parkinson's remedies	can inhibit sweating, lower systolic blood pressure, Cause dizziness and confusion
Antidepressants	inhibit sweating, can inhibit central temperature regulation and limit cognitive alertness
Anxiolytics and muscle relaxants	inhibit sweating, increase dizziness, reduce Cardiac output and thus cooling through vasodilation and worsen breathing
Antiadrenergics and beta-blockers	can prevent the blood vessels in the skin from expanding and thus reduce the ability to dissipate heat through convection
Sympathomimetics	Vasodilators may worsen hypotension in vulnerable patients
Antihypertensives and diuretics	can lead to dehydration and lower blood pressure. Hyponatremia is a common side effect that can be exacerbated by excessive fluid intake
Antiepileptic drugs	can limit cognitive alertness and increase dizziness
other drug groups such as antiemetics, Medicines for dizziness and urinary incontinence as well as gastrointestinal medications	
	also have anticholinergic properties

Possible consequences of excessive heat stress

- increased cardiac output (for cooling through vasodilation)
- Dehydration, altered blood volume distribution and temperature regulation as well as altered drug levels with consequences for kinetics and excretion and thus also the respective pharmacological Activity (all medications)
- increased toxicity or reduced efficacy of medicinal products, especially in cases of narrow therapeutic Index (digoxin or lithium)

Depending on the regulations, **medicines** must be stored and transported at temperatures **below 25°C** or in a refrigerator. High temperatures can reduce the effectiveness of medicines, which are usually only approved for storage up to 25°C. This is especially true for emergency medicines such as antibiotics, adrenergics, insulin, analgesics and sedatives.

The **contents of emergency kits** heat up in warm weather (Crichton, 2004). Healthcare professionals should be careful not to transport medications in heat for long periods of time and to store emergency kits in a cool place.

Many of the above-mentioned conditions require medication, which can increase the risk of heat-related health effects. Rather than forgoing essential medication, patients should be given access to cool rooms so that they are not exposed to the heat.

Antipyretics are ineffective against a heat-related increase in body temperature. They only reduce body temperature if it has increased due to pyrogens. Their use to treat heat-related illnesses can be harmful to the kidneys and liver due to side effects.



Many medicines can
Diarrhea and vomiting
and thus increase the risk
of dehydration in hot
weather.



Mild and moderate heat-related illnesses and their treatment:

Vocabulary

Small red itchy papules on the face, neck, chest, groin and testicles. This can affect people of all ages but is particularly common among young children. Staphylococcus infections can occur. It is associated with excessive sweating in hot, humid weather.

Treatment: Heat rash resolves without special treatment. Minimize sweating by staying in an air-conditioned environment, showering frequently, and wearing light clothing. Keep the affected area dry. Topical antihistamines and antiseptic preparations can be used to reduce discomfort and prevent secondary infection.

Heat edema

Edema in the lower legs, usually at the ankles, at the beginning of the hot season. The phenomenon is attributed to heat-induced peripheral vasodilation and the retention of water and salt.

Treatment: Treatment is not necessary as the edema usually subsides after acclimatization. Diuretics are not recommended.

Heat fainting

Brief loss of consciousness or dizziness when standing upright. This often occurs in patients with cardiovascular disease or in people taking diuretics before acclimatization. The phenomenon is attributed to dehydration, peripheral vasodilation and reduced venous return with reduced cardiac output.

Treatment: The patient should rest in a cool place in a supine position with the legs and hips elevated to increase venous return. Other serious causes for fainting must first be ruled out.

Heat cramps

Painful muscle cramps, usually in the legs, arms or abdomen, often after excessive physical exertion. This phenomenon is attributed to dehydration, electrolyte loss through excessive sweating and muscle fatigue.

Treatment: Immediate rest in a cool place. Stretch and gently massage muscles. Oral rehydration with an electrolyte solution may be necessary. Medical attention should be sought if heat cramps last longer than one hour.

Heat exhaustion

Symptoms include intense thirst, weakness, malaise, tightness, dizziness, fainting and headache. The core temperature may be normal, below normal or slightly elevated (below 40°C). Thready pulse with drop in blood pressure and rapid, shallow breathing. The mental state is unchanged. This phenomenon is associated with water or Salt loss due to exposure to high ambient heat or strenuous physical activity.

Treatment: Move the patient to a cool, shaded room or air-conditioned area. Clothing should be removed. Apply cold, wet towels or spray cold water and use a fan if one is available. Place the patient on his/her back with the legs and hips elevated to increase venous return. Initiate oral hydration. If nausea prevents oral intake of fluids, consider intravenous hydration. If hyperthermia exceeds 39°C, confusion, or persistent hypotension occurs, treat as heat stroke and transport to hospital.

Treatment of life-threatening heat stroke:

PHYSICAL

CONDITION

MEASURE

GOAL

Outpatient treatment

Heat overload (heat wave, summer heat or strenuous activity)

Measure core temperature (rectally). If $> 40^{\circ}\text{C}$, move patient to a cooler place, remove clothing, cool externally. Place cold packs on neck, armpits and groin, fan constantly (or leave window open) and spray skin with water at $25 - 30^{\circ}\text{C}$.

Diagnose heat stroke. Reduce core temperature to $< 39.4^{\circ}\text{C}$. Promote cooling by dissipation, maintain airflow. Promote cooling by evaporation.

Changes in mental state (anxiety, Delirium, epileptic Seizures, coma)

Place the unconscious person in the recovery position with a clear airway. Give oxygen (4 l/min), isotonic crystalloid (normal saline solution). Transfer to the emergency room.

Minimize the risk of aspiration. Arterial oxygen saturation $> 90\%$. Ensure volume expansion.

Inpatient treatment

Hyperthermia

Confirm diagnosis with a thermometer calibrated to measure high temperatures ($40 - 47^{\circ}\text{C}$). Monitor skin and rectal temperature, and continuously cool.

Goal: Maintain skin temperature at $> 30^{\circ}\text{C}$. Stop cooling when rectal temperature is $< 39.4^{\circ}\text{C}$.

Epileptic seizures

Consider administration of benzodiazepines.

Prevent epileptic seizures.

Respiratory failure

Consider elective intubation (if gag and cough reflexes are impaired or breathing is impaired).

Protect the airways and increase oxygen administration (arterial oxygen saturation to $> 90\%$).

Hypotension

Administer volume expanders, add vasoconstrictors, and consider monitoring central venous pressure.

Increase mean arterial pressure > 60 mm Hg, restore organ perfusion and tissue oxygenation (consciousness, 24-hour urine collection, lactate level).

Rhabdomyolysis

Expand volume with normal saline, give intravenous furosemide and mannitol or intravenous sodium bicarbonate. Monitor serum potassium and calcium and treat even mild hyperkalemia.

Prevent kidney damage induced by myoglobin. Promote renal blood flow and diuresis. Ensure urine alkalization.

Dysfunction of multiple organs

Nonspecific supportive therapy.

Support the recovery of organ function.

After cooling

Prevent life-threatening cardiac arrhythmias.

3.2.6. Medical professionals and nursing services

Chronic, but not acute, diseases are taken into account. However, infections, fever, gastrointestinal catarrh and skin infections are also risk factors for heat-related mortality:

ILLNESS

IMPACT

Diabetes mellitus and other endocrine Disturbances

Type 1 and 2 diabetes are associated with impaired blood circulation in the skin, which can lead to reduced heat dissipation. Metabolism and sweat function can also be impaired

Organic mental disorders, dementia, Alzheimer's disease

Reduced awareness of heat-related risks and behavioral adjustments, great need for assistance, interaction of several drugs in regulating body temperature

Disturbances due to Drug abuse

Changes in physiological responses and behavior due to psychoactive substances and alcohol

Schizophrenia, schizotypal and delusional disorders

Great need for assistance, prescription of psychotropic medication

Neurological diseases, e.g. Parkinson's disease and diseases associated with cognitive disorders

Potentially impaired consciousness and mobility, high need for care, prescription of psychotropic medication

Cardiovascular diseases (including High blood pressure, coronary heart disease, Conduction disorders)

impaired regulation of body temperature and high risk of acute coronary artery and cerebral thrombosis, reduced circulatory response and temperature regulation as well as altered blood composition due to dehydration (deficit of 1% of body weight). Changes in renal function can lead to life-threatening cardiac arrhythmias in elderly patients. Deterioration due to changes in circulatory function, temperature regulation and blood composition can lead to fatal cerebral ischemia in hypertensive patients after a sudden drop in arterial pressure. Changes in the peripheral circulation can lead to poorer regulation of core body temperature.

Respiratory diseases, chronic Lower respiratory tract disease

The combination of heat and air pollution influences the pathogenesis and clinical history of respiratory diseases (asthma, chronic bronchitis); worsening of existing diseases (chronic obstructive pulmonary disease (COPD)) due to hyperventilation and dyspnea; difficulty in heat dissipation (dilation of peripheral vessels, hypovolemia).

Kidney disease, kidney failure, Kidney stones

reduced renal function due to electrolyte and water imbalance resulting from hyperthermia and dehydration, particularly in the elderly.

Obesity

impaired heat perception or reduced ability to dissipate heat due to a smaller ratio of body surface to body mass, which hinders sweat evaporation

Other chronic diseases

Sweat gland loss due to scleroderma, in cystic fibrosis high Electrolyte loss through sweating

Anyone suffering from any of the above-mentioned conditions should seek medical advice before the start of summer about what to consider during the warm season.

3.2.7. Municipalities

Cool places for the population to relax and cool down during a heat wave:

- Community centers
- Cultural centers and museums
- Cinemas
- Churches
- Libraries
- Public swimming pools
- Local recreation areas
- Shopping centers

Adjusting opening hours should be considered if necessary and a transport system for immobile people should be installed or tested. The distribution of drinking water and/or the installation of drinking fountains in public buildings should be emphasized.

3.2.8. Care facilities for children and young people

- Give small portions to drink throughout the day (avoid sugary drinks)
- Serve fresh, cool, light food (melons, cucumbers, fruits, etc.)
- Play in forests or parks with dense trees (temperature and ozone levels lower)
- Protect yourself from heat and sunlight (light clothing, hat, stay in the shade, sunscreen)
- In consultation with parents, identify those persons who are particularly at risk in hot weather (children with chronic diseases)
- To prevent diarrhoea, hygienic handling of food
- Install drinking water dispenser
- Do not leave children in parked cars or alone in enclosed spaces
- Move sports and gymnastics events to the morning
- Appoint responsible persons and appropriately trained educators
- Establish organizational procedures for emergencies

3.2.9. Hospital management

During longer periods of heat, internal resources may become overloaded – so it is important to correctly assess the capacity of your own business in order to be able to plan and implement adaptation measures:

- Creation of an in-house action plan for heat stress
- Identification of patients at risk
- Analysis of the spatial conditions, planning and implementation of necessary measures such as External blinds, ventilation options, air conditioning
- Ensuring the flow of information in the house •

Training/retraining of care staff

- Light bedding
- Lightweight, breathable workwear
- Install drinking water dispensers
- Human resource planning

3.2.10. Emergency and blue light organisations

When heat waves are approaching, emergency and rescue organizations are warned directly so that they can plan and implement adaptation measures.

- Develop an action plan for heat stress
- Identification of patients at risk
- Ensuring the flow of information •

Training/retraining of care staff • Light, breathable

work clothing (if permitted under labor law) • Setting up drinking water dispensers

- Human resource planning



4. Emergency numbers, Internet and contact persons

4.1. General emergency numbers

Telephone health advice	1450	Emergency medical service	141
Pharmacies	1455	police	133
rescue	144	Emergency call for the deaf: fax or SMS	0800 133 133

4.2. Internet

Emergency medical service	www.141-vorarlberg.at
Z.A.M.G.	http://warnungen.zamg.ac.at
Wetterbox Vorarlberg	http://niederschlag.lwz-vorarlberg.at/

Country warning homepage

www.vorarlberg.at/warnung | <http://warnung.vorarlberg.at>

Vorarlberg State Warning Center

@lwzvorarlberg

 lwzvorarlberg

Public health portal of Austria www.gesundheit.gv.at/aktuelles/archiv-2017/hitzeschutzplan

Ozon-Information: <http://www.vorarlberg-luft.at/>

Ozontelefon: 05574 511-42973

4.3. Contact person

4.3.1. Bund

AGES Talkphone

+43 800 555 621

4.3.2. Office of the Vorarlberg State Government

Department of Health Affairs

Römerstrasse 15

6900 Bregenz

T +43 5574 511 24405

4.3.3. District administrative authority

Health Department, BH Bludenz

Schloss-Gayenhof-Platz 2, 6700 Bludenz
T +43 5552 6136 51614

Health Department, BH Bregenz

Bahnhofstrasse 41, 6900 Bregenz
T +43 5574 4951 52615

Health Department, Dornbirn District Council

Klaudiastrasse 2, 6850 Dornbirn
T +43 5572 308 53614

Health Department, Feldkirch District Council

Schloßgraben 1, 6800 Feldkirch
T +43 5522 3591 54617

4.3.4. Hospitals

Bludenz State Hospital

Spitalgasse 3, 6700 Bludenz
T +43 5552 60303

Bregenz State Hospital

Carl-Pedenz-Strasse 12, 6900 Bregenz
T +43 5574 4011

Feldkirch State Hospital

Carinagasse 47, 6800 Feldkirch
T +43 5522 303

Hohenems State Hospital

Bahnhofstrasse 31, 6845 Hohenems
T +43 5576 7033



VORARLBERGER KRANKENHAUS-BETRIEBSGES.M.B.H.

Dornbirn Hospital

Lustenauer Str. 4, 6850 Dornbirn T
+43 5572 3030



4.3.5. State associations for outpatient care and nursing

Mobile Aid Service Association Dr.-Waibel-Straße
3, 6850 Dornbirn T +43 5572 398297



State Association of Home Care Dr.-
Waibel-Straße 3, 6850 Dornbirn T +43
5572 34935



Department of Health Affairs

Römerstrasse 15

6900 Bregenz

T +43 5574 511 24405

5. Targeted heat warnings

In addition to the information services (homepage) of the ZAMG, the Federal Ministry, the Office of the Vorarlberg State Government and the AGES heat hotline, the various institutions in the state of Vorarlberg are informed via the media, email, SMS or telephone with tips and behavioral measures on impending heat waves from the heat protection plan. These include care and nursing facilities, hospitals, mobile assistance services, childcare facilities, schools, municipalities, medical associations, emergency organizations, religious communities, sports clubs and public swimming pools.

The state of Vorarlberg informs the media about an impending heat stress by the Central Institute for Meteorology and Geodynamics (ZAMG) as follows:

Ladies and Gentlemen!

Due to the impending heat wave, we refer to the heat protection plan available on the Vorarlberg state website (**www.vorarlberg.at**) with the request that the population and affected institutions be informed accordingly.





6. Information material, brochures

“Healthy despite the

heat” www.umweltbundesamt.at/aktuelles/presse/news2019/news20190613

“Summer and heat”

www.sozialministerium.at

“Stay cool”

Safe Vorarlberg in cooperation with KLAR! In Walgau



The heat protection plan is available for download on the Internet at **www.vorarlberg.at/hitzeschutzplan** and **www.xsund.at** as well as in the **Xsund app** .

Attachment

Sources •

Austrian Public Health Portal • Heat-
Health Action Plans WHO/Europe

- Health advice on preventing heat-related health damage WHO/Europe
- Federal Environment Agency
- Climate maps Vorarlberg
- Hessian Action Plan to Prevent Heat-Related Health Impairments of the Population (HEAT)
- Styrian heat protection plan
- Parliamentary inquiry and response dated 19 July 2019
- Vienna heat guide
- Collection of guidelines for childcare facilities and playgroups in the state of Vorarlberg (LeiLa)



Office of the Vorarlberg State Government

Department of Health and Sport

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gesundheitundsport@vorarlberg.at

www.vorarlberg.at/hitzeschutzplan